Linear Algebra – HW 1 Solution

Fuhan 10245501401

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Problem 1 Find the inverse of the matrix

$$A := \begin{bmatrix} 2 & 1 & 3 \\ 0 & 1 & 2 \\ 1 & 0 & 3 \end{bmatrix}.$$

Solution. Using the adjugate formula, we obtain

$$A^{-1} = \frac{1}{\det A} A^* = \frac{1}{5} \begin{bmatrix} 3 & -3 & 1 \\ 2 & 3 & -4 \\ -1 & 1 & 2 \end{bmatrix}.$$

Problem 2 Solve the linear system

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 4 \end{bmatrix}. \tag{2.1}$$

Solution. Apply Gaussian elimination to the augmented matrix associated with Eq (2.1):

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & 1 & 4 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Therefore, the solution is $(x, y, z) = (1, 1, 1)$.

Problem 3 Suppose $\alpha_1 = (1, 3, 4, -2)^\top$, $\alpha_2 = (2, 1, 3, t)^\top$, and $\alpha_3 = (3, -1, 2, 0)^\top$ are linearly dependent. Find t.

Solution. Consider the matrix

$$A = [\alpha_1 \ \alpha_2 \ \alpha_3].$$

The vectors α_1 and α_3 are linearly independent, so $\operatorname{rank} A \geq 2$. Since $\alpha_1, \alpha_2, \alpha_3$ are linearly dependent, $\operatorname{rank} A < 3$. Hence $\operatorname{rank} A = 2$. The rank equals the order of the largest nonzero minor, so

$$\det \begin{bmatrix} 3 & 1 & -1 \\ 4 & 3 & 2 \\ -2 & t & 0 \end{bmatrix} = 0 \Rightarrow t = -1.$$

Therefore, $t = -1$.

Problem 4 Let $\alpha_1, \alpha_2, \alpha_3$ be linearly independent. Prove that $\alpha_1 + \alpha_2$, $\alpha_2 + \alpha_3$, and $\alpha_1 + \alpha_3$ are also linearly independent.

Proof. Suppose

$$\begin{aligned} & k_1(\alpha_1 + \alpha_2) + k_2(\alpha_2 + \alpha_3) + k_3(\alpha_1 + \alpha_3) = \mathbf{0}. \\ \Rightarrow \ & (k_1 + k_3)\alpha_1 + (k_1 + k_2)\alpha_2 + (k_2 + k_3)\alpha_3 = \mathbf{0} \end{aligned} \tag{4.1}$$

Since $\alpha_1, \alpha_2, \alpha_3$ are linearly independent, Eq (4.1) implies $k_1 + k_3 = k_1 + k_2 = k_2 + k_3 = 0$. Thus $k_1 = k_2 = k_3 = 0$, proving the result. ■

Problem 5 Find a basis for the following set of vectors, and express each remaining vector as a linear combination of that basis:

$$\begin{aligned}\alpha_1 &= (1,2,-1,3)^\top,\\ \alpha_2 &= (2,4,-2,6)^\top,\\ \alpha_3 &= (1,0,1,1)^\top,\\ \alpha_4 &= (3,2,1,5)^\top.\end{aligned}$$

Solution. We have $\alpha_2 = 2\alpha_1$ and $\alpha_4 = \alpha_1 + 2\alpha_3$. Moreover, α_1 and α_3 are linearly independent. Therefore, $\{\alpha_1, \alpha_3\}$ is a basis for the set.

Problem 6 Let $\varphi : \mathbb{R}^3 \to \mathbb{R}^4$ be the linear map

$$\varphi\left(\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}\right) = \begin{bmatrix}3x_1 + 2x_2 + x_3\\ x_1 + x_2 + x_3\\ x_1 - 3x_2\\ 2x_1 + 3x_2 + x_3\end{bmatrix}.$$

Use the standard bases e_1, e_2, e_3 of $\mathbb{R}^3$ and $\hat{e_1}, \hat{e_2}, \hat{e_3}, \hat{e_4}$ of $\mathbb{R}^4$.

(1) Compute the matrix $\boldsymbol{A}_\varphi$.
(2) Compute $\operatorname{rank} \boldsymbol{A}_\varphi$.

Solution.
(1) By definition,

$$\varphi[e_1\ \ e_2\ \ e_3] = \begin{bmatrix}3 & 2 & 1\\ 1 & 1 & 1\\ 1 & -3 & 0\\ 2 & 3 & 1\end{bmatrix} = [\hat{e_1}\ \ \hat{e_2}\ \ \hat{e_3}\ \ \hat{e_4}]\begin{bmatrix}3 & 2 & 1\\ 1 & 1 & 1\\ 1 & -3 & 0\\ 2 & 3 & 1\end{bmatrix}.$$

Hence

$$\boldsymbol{A}_\varphi = \begin{bmatrix}3 & 2 & 1\\ 1 & 1 & 1\\ 1 & -3 & 0\\ 2 & 3 & 1\end{bmatrix}.$$

(2) Elementary row operations give

$$\boldsymbol{A}_\varphi \sim \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}.$$

Therefore, $\operatorname{rank} \boldsymbol{A}_\varphi = 3$.

Problem 7 Let $\mathcal{A} \in \operatorname{End}(\mathbb{R}^3)$ be defined by

$$\begin{cases}\mathcal{A}(\eta_1) = (-5,0,3)^\top\\ \mathcal{A}(\eta_2) = (0,-1,6)^\top\\ \mathcal{A}(\eta_3) = (-5,-1,9)^\top\end{cases}$$

where

$$\begin{cases}\eta_1 = (-1,0,2)^\top\\ \eta_2 = (0,1,1)^\top\\ \eta_3 = (3,-1,0)^\top.\end{cases}$$

(1) Find the matrix of $\mathcal{A}$ with respect to the standard basis e_1, e_2, e_3.
(2) Find the matrix of $\mathcal{A}$ with respect to the basis η_1, η_2, η_3.

Solution.

(1) From the definition,

$$\begin{aligned}\mathcal{A}[\eta_1 \ \eta_2 \ \eta_3] &= [e_1 \ e_2 \ e_3]P,\\ [\eta_1 \ \eta_2 \ \eta_3] &= [e_1 \ e_2 \ e_3]S,\end{aligned}$$

where

$$P = \begin{bmatrix}-5 & 0 & -5\\ 0 & -1 & -1\\ 3 & 6 & 9\end{bmatrix}, \quad S = \begin{bmatrix}-1 & 0 & 3\\ 0 & 1 & -1\\ 2 & 1 & 0\end{bmatrix}.$$

It follows that

$$\begin{aligned}&\mathcal{A}[e_1 \ e_2 \ e_3]S = [e_1 \ e_2 \ e_3]P,\\ \Rightarrow \quad &\mathcal{A}[e_1 \ e_2 \ e_3] = [e_1 \ e_2 \ e_3]PS^{-1}.\end{aligned}$$

Therefore, the matrix of $\mathcal{A}$ in the standard basis is

$$A := PS^{-1} = \frac{1}{7}\begin{bmatrix}-5 & 20 & -20\\ -4 & -5 & -2\\ 27 & 18 & 24\end{bmatrix}.$$

(2) Since

$$\begin{aligned}&\mathcal{A}[e_1 \ e_2 \ e_3] = [e_1 \ e_2 \ e_3]A,\\ \Rightarrow\ &\mathcal{A}[\eta_1 \ \eta_2 \ \eta_3] = [\eta_1 \ \eta_2 \ \eta_3]S^{-1}AS,\end{aligned}$$

the matrix in the basis η_1, η_2, η_3 is

$$S^{-1}AS = \begin{bmatrix}2 & 3 & 5\\ -1 & 0 & -1\\ -1 & 1 & 0\end{bmatrix}.$$